Pooled R², cruise held out

		Granite embedding	In-situ temperature	Difference (G − T)	Main carrier, share of genes (one example metagenome)
Granite clearly better				28 shown of 360 KOs	
Transport	**pstS** phosphate transport system substrate-binding…	0.46	0.25	+0.22	Pseudomonadota 59%
	pstB phosphate transport system ATP-binding protein	0.46	0.17	+0.29	Pseudomonadota 68%
	pstA phosphate transport system permease protein	0.42	0.16	+0.26	Pseudomonadota 60%
	pstC phosphate transport system permease protein	0.39	0.08	+0.32	Pseudomonadota 69%
	thiY putative hydroxymethylpyrimidine transport s…	0.38	0.06	+0.32	Pseudomonadota 84%
Amino acids	**pheA2** prephenate dehydratase	0.47	0.20	+0.27	Pseudomonadota 43%
	glsA glutaminase	0.46	0.17	+0.29	Eukaryotes 35%
	E2.5.1.54 3-deoxy-7-phosphoheptulonate synthase	0.44	0.16	+0.28	Pseudomonadota 46%
	LACC1 purine-nucleoside/S-methyl-5'-thioadenosine…	0.44	0.23	+0.20	Pseudomonadota 50%
	lysA diaminopimelate decarboxylase	0.43	0.06	+0.37	Pseudomonadota 49%
	argH argininosuccinate lyase	0.42	0.10	+0.32	Pseudomonadota 49%
	argG argininosuccinate synthase	0.40	0.06	+0.35	Pseudomonadota 46%
	proA glutamate-5-semialdehyde dehydrogenase	0.40	0.01	+0.39	Pseudomonadota 55%
Nucleotides	**PFAS** phosphoribosylformylglycinamidine synthase	0.46	0.22	+0.24	Pseudomonadota 44%
	purK 5-(carboxyamino)imidazole ribonucleotide syn…	0.42	0.16	+0.26	Pseudomonadota 41%
Carbohydrate, lipid & cofactors	**glcF** glycolate dehydrogenase iron-sulfur subunit	0.45	0.23	+0.21	Pseudomonadota 57%
	E4.1.3.25 (S)-citramalyl-CoA lyase	0.44	0.16	+0.28	Pseudomonadota 83%
	nadA quinolinate synthase	0.40	0.12	+0.28	Pseudomonadota 41%
	dhaA haloalkane dehalogenase	0.40	0.25	+0.15	Pseudomonadota 63%
	ispH 4-hydroxy-3-methylbut-2-en-1-yl diphosphate…	0.40	0.15	+0.25	Pseudomonadota 49%
	ENO1_2_3 enolase 1/2/3	0.40	0.20	+0.20	Pseudomonadota 46%
	pntA proton-translocating NAD(P)+ transhydrogenas…	0.39	0.24	+0.15	Pseudomonadota 51%
DNA, RNA & protein synthesis	**scpA** segregation and condensation protein A	0.48	0.27	+0.21	Pseudomonadota 49%
	FARSA phenylalanyl-tRNA synthetase alpha chain	0.41	0.18	+0.23	Pseudomonadota 42%
	udg uracil-DNA glycosylase	0.40	0.15	+0.25	Pseudomonadota 53%
	CARS cysteinyl-tRNA synthetase	0.39	0.16	+0.24	Pseudomonadota 43%
Other / uncharacterised	**tldD** TldD protein	0.43	0.25	+0.18	Pseudomonadota 39%
	K06893 uncharacterised protein	0.42	0.13	+0.29	Pseudomonadota 70%
Both high, about equal				14 shown of 37 KOs	
Energy & photosynthesis	**hoxH** NAD-reducing hydrogenase large subunit	0.71	0.75	-0.04	Cyanobacteria 99%
	ndhD NAD(P)H-quinone oxidoreductase subunit 4	0.71	0.76	-0.05	Cyanobacteria 98%
	ndhK NAD(P)H-quinone oxidoreductase subunit K	0.70	0.70	-0.01	Cyanobacteria 100%
	ndhJ NAD(P)H-quinone oxidoreductase subunit J	0.69	0.73	-0.04	Cyanobacteria 90%
	cpeS phycoerythrin-associated linker protein	0.67	0.68	-0.01	Cyanobacteria 99%
Transport	**mntB** manganese transport system permease protein	0.70	0.69	+0.00	Cyanobacteria 66%
	mntC manganese transport system substrate-binding…	0.69	0.66	+0.04	Cyanobacteria 100%
	ecfT energy-coupling factor transport system perm…	0.67	0.67	-0.00	Cyanobacteria 41%
Nucleotides	**nrdJ** ribonucleotide reductase, class II	0.67	0.68	-0.02	Cyanobacteria 100%
Carbohydrate, lipid & cofactors	**K14330** fatty aldehyde-generating acyl-ACP reductase	0.69	0.71	-0.02	Cyanobacteria 100%
	chlL light-independent protochlorophyllide reduct…	0.65	0.69	-0.04	Pseudomonadota 65%
	E3.1.2.28 1,4-dihydroxy-2-naphthoyl-CoA hydrolase	0.65	0.67	-0.03	Cyanobacteria 78%
Other / uncharacterised	**K07015** Rab family protein	0.67	0.70	-0.03	Cyanobacteria 100%
	mviM virulence factor MviM	0.67	0.68	-0.01	Cyanobacteria 59%
Temperature clearly better				12 shown of 280 KOs	
Energy & photosynthesis	**ndhF** NAD(P)H-quinone oxidoreductase subunit 5	0.53	0.65	-0.12	Cyanobacteria 95%
	puhA photosynthetic reaction center H subunit	0.31	0.44	-0.13	Pseudomonadota 74%
	pufL photosynthetic reaction center L subunit	0.31	0.44	-0.13	Pseudomonadota 83%
Transport	**fiu** catecholate siderophore receptor	0.32	0.51	-0.20	Verrucomicrobiota 58%
Nucleotides	**RRM2** ribonucleoside-diphosphate reductase subunit…	0.27	0.40	-0.13	dsDNA viruses, no RNA stage 44%
Carbohydrate, lipid & cofactors	**dgkA** diacylglycerol kinase (ATP)	0.45	0.57	-0.12	Eukaryotes 44%
	GPAT3_4 glycerol-3-phosphate O-acyltransferase 3/4	0.34	0.45	-0.11	Eukaryotes 99%
	ams amylosucrase	0.35	0.45	-0.10	Cyanobacteria 88%
DNA, RNA & protein synthesis	**rex** redox-sensing transcriptional repressor	0.40	0.56	-0.16	Planctomycetota 34%
	POLD1 DNA polymerase delta subunit 1	0.36	0.47	-0.11	dsDNA viruses, no RNA stage 50%
	TOP2 DNA topoisomerase II	0.23	0.44	-0.21	dsDNA viruses, no RNA stage 41%
	NOP2 25S rRNA (cytosine2870-C5)-methyltransferase	0.29	0.39	-0.10	Eukaryotes 96%

≤0 0.4 0.8
R²

T 0 Granite
ΔR², better:

Best model so far: ridge regression on 32 principal components of the OSTIA-filled Granite embedding + in-situ temperature (median pooled R² 0.34 on the 20-KO benchmark, cruise held out; RESULTS.md R12). The heatmap compares the embedding alone (ridge) with temperature alone (spline).